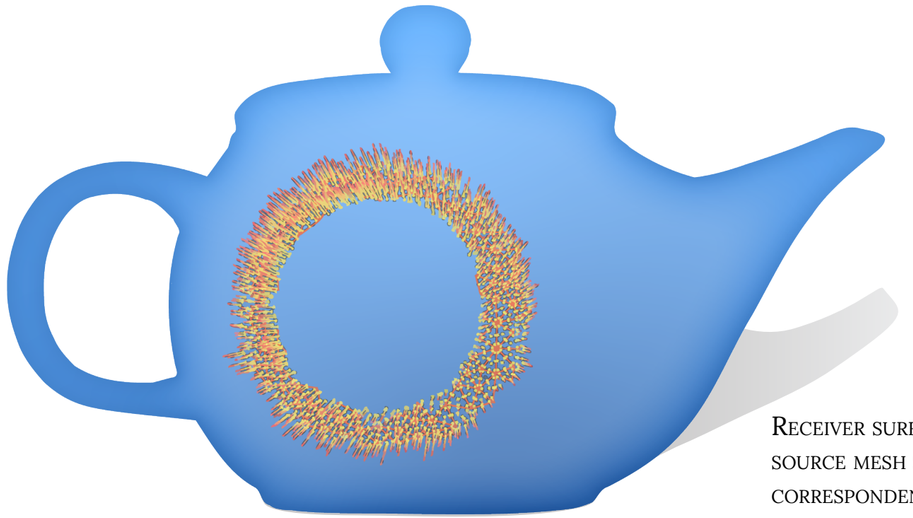
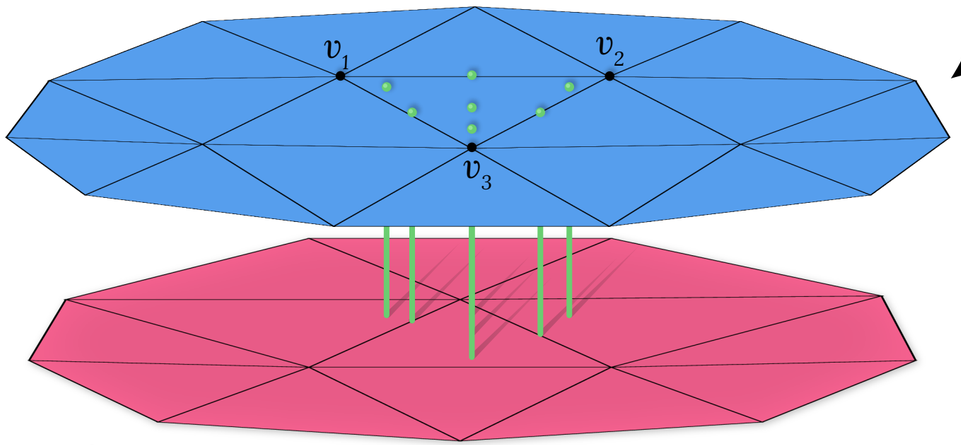


Receiver surface mesh (teapot) sampling source mesh (torus). Lines represent correspondence mapping

Contact area is determined by 7 quadrature points. Each point originating from the receiver surface samples the surface of a source mesh to determine contact correspondence.

Receiver Surface



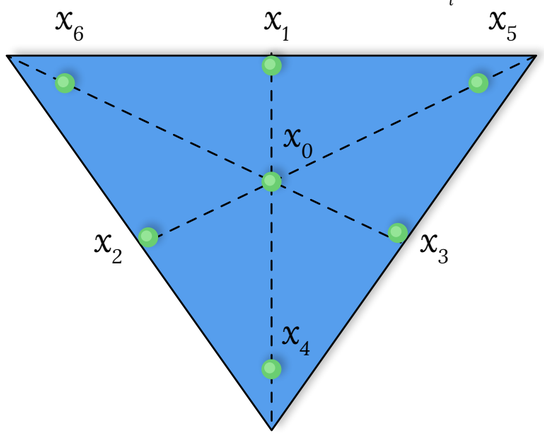
Contact correspondence

Source Surface

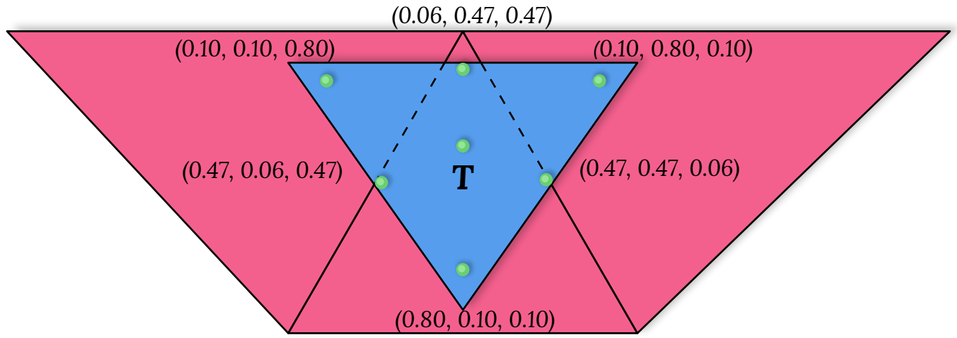
Barycentric Coordinate: $\mathbf{b}=(b_1,b_2,b_3)$

$$x(\mathbf{b})=b_1\mathbf{v}_1+b_2\mathbf{v}_2+b_3\mathbf{v}_3$$

x_i is the local position of a quadrature point and has an associated weight w_i



The associated weight is fractional area and is considered in the final numerical integration only if correspondence has been established. Otherwise, it is not considered.

$$\text{Area}(\mathbf{T}) = \tfrac{1}{2} \cdot |\, det[\, \mathbf{v}_2 - \mathbf{v}_1,\ \mathbf{v}_3 - \mathbf{v}_1 \,] = \tfrac{1}{2} \cdot \|\, (\mathbf{v}_2 - \mathbf{v}_1) \times (\mathbf{v}_3 - \mathbf{v}_1) \,\|$$

$$\iint_T f(x)\, dA \approx Area(\mathbf{T}) \cdot \sum_{i=1}^{7} w_i\, f(x(\mathbf{b}^{(i)}))$$